

SIJAN PANDEY

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Education

Kathmandu University

Kavre, Dhulikhel, Nepal

Bachelor of Technology in Artificial Intelligence(7th Sem)

2022 – Present

- Coursework: Deep Learning · Machine Learning · Computer Vision · NLP · LLMs · Graph Neural Networks · GIS & Remote Sensing · Agentic AI Systems

Tilottama Multiple Campus

Rupandehi, Nepal

Higher Secondary Education (+2), Science

2019 – 2022

Technical Skills

Languages: Python, C/C++, Java, JavaScript, Dart, SQL

ML & AI: PyTorch, TensorFlow/Keras, YOLOv8, OpenCV, Dlib, Scikit-learn, Hugging Face, FAISS, LangGraph

Web & Backend: FastAPI, Flutter, React, Docker, MongoDB

Tools: Git, QGIS, Blender, Jupyter, Linux

Domains: Computer Vision · Face Recognition · NLP · GIS & Remote Sensing · Agentic AI · IoT

Projects

Landslide Susceptibility Mapping – Nepal | QGIS · Python · rasterio · Scikit-learn

2025

- Produced ML-based susceptibility maps of Nepal landslide risk zones by fusing multi-band Landsat composites, DEM derivatives, and categorical geo-features into a supervised classification pipeline.

Reaco – AI-Powered Photo Distribution for Photographers | YOLOv8 · Siamese Net · FastAPI · Flutter · Docker

2024

- Built an end-to-end automated photo distribution system for photographers: YOLOv8m detects and crops faces from bulk uploads, Dlib aligns them, and an InceptionV3 Siamese network trained with triplet loss generates 128-d embeddings for N-shot KNN identity clustering.
- Assembled a training corpus of 220,470+ face images across 11,197 individuals; deployed on FastAPI + MongoDB + Docker with a Flutter/Dart mobile interface for event-based photo retrieval.

Open Set Facial Recognition & Surveillance | PyTorch · TensorFlow · FastAPI · Electron · MongoDB

2024

- Developed an open-set surveillance system using CNNs, N-shot learning, and KNN to identify registered individuals while correctly flagging unknown faces in real-time footage.
- Built FastAPI backend for inference serving, Electron desktop client for operator control, and MongoDB for large-scale identity and event data storage.

IoT Hydroponics Monitoring System | ESP32 · KiCad · C/C++

2024

- Developed an AI-powered hydroponic system for lettuce cultivation integrating IoT-based monitoring of pH, EC, temperature, humidity, water level, and light intensity. Implemented anomaly detection, computer vision-based plant health monitoring, and automated control using ESP32, Raspberry Pi, FastAPI, MongoDB, MQTT, and Flutter..

Nepali Food Classifier | Python · TensorFlow · Keras · CNN · Flutter · Dart

2023

- Integrated YOLOv8, InceptionV3, and VGG16 models for traditional Nepali food categories using TensorFlow/Keras; assembled a custom dataset of 5,000 images with over 90% accuracy.

Experience & Leadership

Kathmandu University AI Club (KUAIC)

Dhulikhel, Nepal

AI Conclave Coordinator

Aug 2025 – Jan 2026

- Led AI Conclave 2026, Kathmandu University's flagship national AI event, overseeing end-to-end planning and execution of a three-day program connecting students, researchers, industry experts, and policymakers.
- Managed multidisciplinary teams across sponsorship, operations, marketing, technical competitions, project exhibitions, speaker engagement, and event logistics.
- Secured and coordinated partnerships with government, academic, and industry stakeholders, including the Ministry of Education, Science and Technology (MOEST), National AI Center, Nabil Bank, Fusemachines, and WorldLink Communications.
- Organized expert talks, panel discussions, AI summits, robotics competitions, project showcases, and networking activities to strengthen Nepal's AI ecosystem and industry-academia collaboration.